

Trajectory Optimisation for a Steerable Airdrop in Unknown Wind

Guiding packages to a target for relief distribution, with no wind sensor on board.

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80,000

PEOPLE STILL UNREACHED, ASSAM FLOODS, JULY 2026

A cloudburst on 19 July 2026 flooded four districts of Upper Assam. **705,000 people were affected**, roads were gone, and air was the only way in.

Supplies go out by helicopter. Unguided, a payload rides the air mass down: it does not aim, it drifts. **Here it lands 23.9 m off** — in a flood, the wrong side of the water.



SCAN 1 OF 2 · START HERE

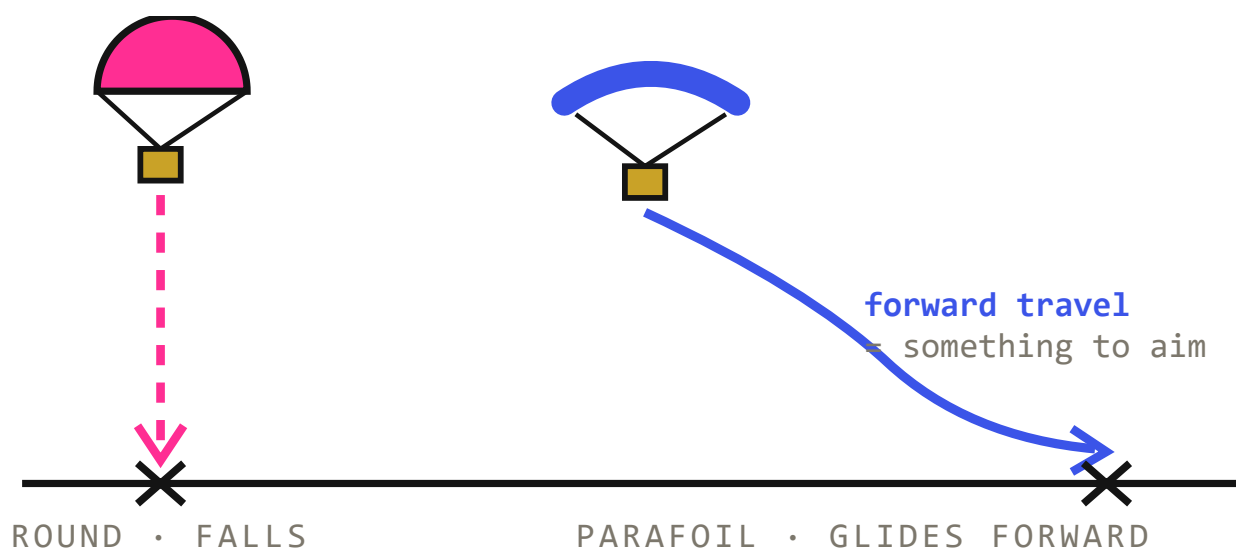
Watch it miss.

The drop above, in your own room. Point your phone at the floor.

ANSWER: YES, IF IT IS A WING

You can steer it. You cannot stop it.

A round parachute only falls. A **parafoil** is a ram-air canopy: it **glides forward** as it descends, so pulling one side turns it. Forward travel is the only thing to aim.



$$v = \sqrt{2mg / \rho S C_L}$$

How fast it flies. Only the wing loading m/S appears — mass and canopy area never act separately.

It has no engine and no brakes. Sink rate is constant, so **the moment it is released its time in the air is fixed** — and with it, the distance it can cover.

That is usually *more* than the distance to the target, and it cannot slow down. **It has to spend the surplus.**

WHAT IT WILL FLY BEFORE IT LANDS



DISTANCE TO THE TARGET



SO → Fly straight at the target and it arrives with height to spare.

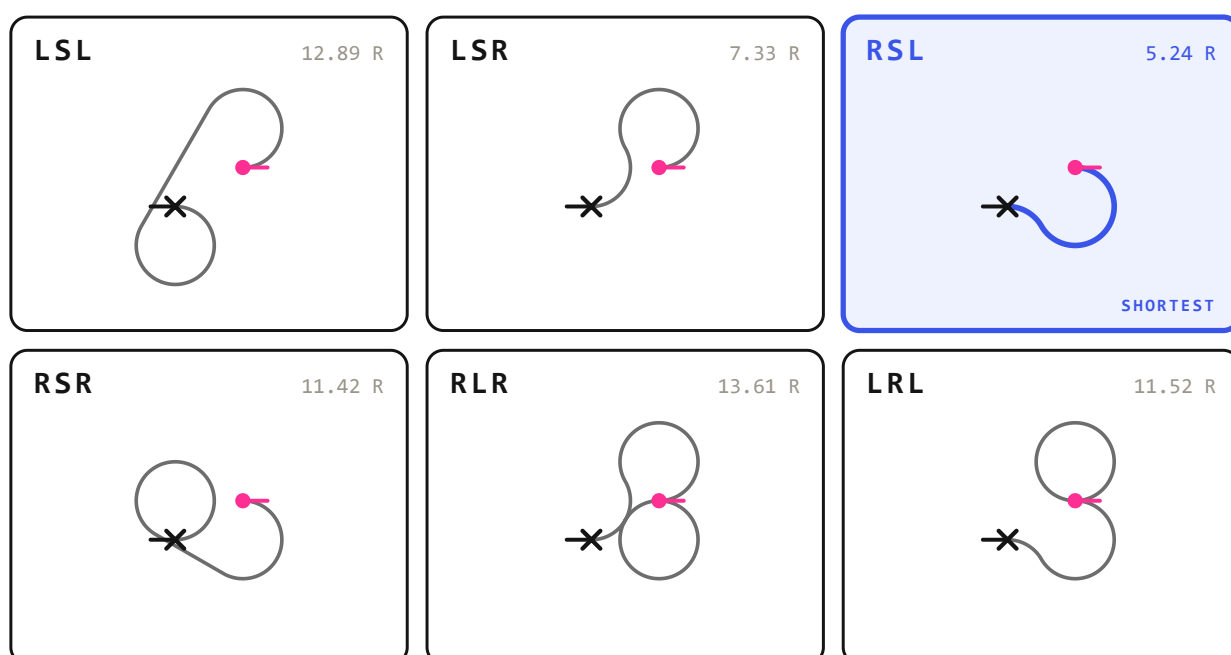
So it weaves — on purpose.

The guidance does not fly the shortest route. It flies one of **the right length**, holding a track angle α off the bearing to stretch the path by $1/\cos \alpha$ until the budget is gone.

$$\cos \alpha = d / s$$

d = distance to go · s = flying left. As height runs out, $d/s \rightarrow 1$ and $\alpha \rightarrow 0$: a straight final approach, no change of mode.

But d is not the straight line. It cannot turn on the spot, so the real distance is a curve with a minimum radius. In **1957 Lester Dubins proved** the shortest such curve is one of six shapes.



Same start, target and headings, solved six ways at one scale. Computed by the model, not drawn. The six closed forms match step-wise integration to 9.0×10^{-12} m over 3,514 solutions.

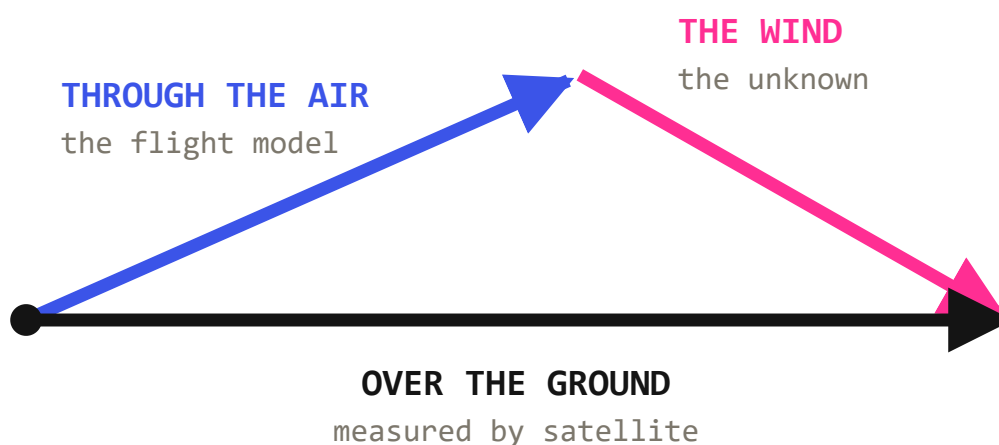
It checks all six, takes the shortest, and measures the budget against *that*. **Dubins measures; the weave spends.**

SO → So it is turning most of the way down.

THE PART THAT PAYS FOR ITSELF

And that is how it finds the wind.

No sensor measures the wind, and nothing is sent from the ground. An **Extended Kalman Filter** works it out from satellite velocity alone, because the wind is a subtraction:



State: $[w_x, w_y, v_a, b]$ – wind east and north, airspeed, magnetometer bias. The bias enters through the heading trigonometry, making the problem non-linear.

Except it should be impossible

On a **fixed heading**, a headwind and a too-low airspeed produce *the same* track. No amount of data separates them — the state is unobservable. Flying straight, this cannot work.

FLYING STRAIGHT

Headwind, or an airspeed that is too low? **Same track.** Unsolvable.

TURNING

Different tracks. The filter can separate them.

And by panel 03 it is *already turning*, for reasons that had nothing to do with the wind. **The measurement is free.**

0.104 m s⁻¹

MEDIAN WIND ERROR RECOVERED · 95% CI 0.096–0.112

BUT → Does it land closer?

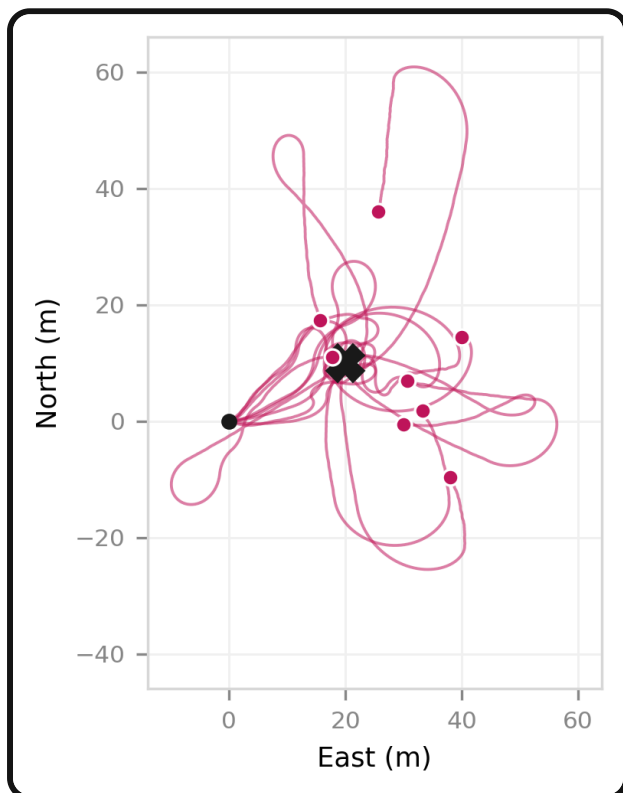
150 RANDOMISED DROPS

Yes — less than half the miss.

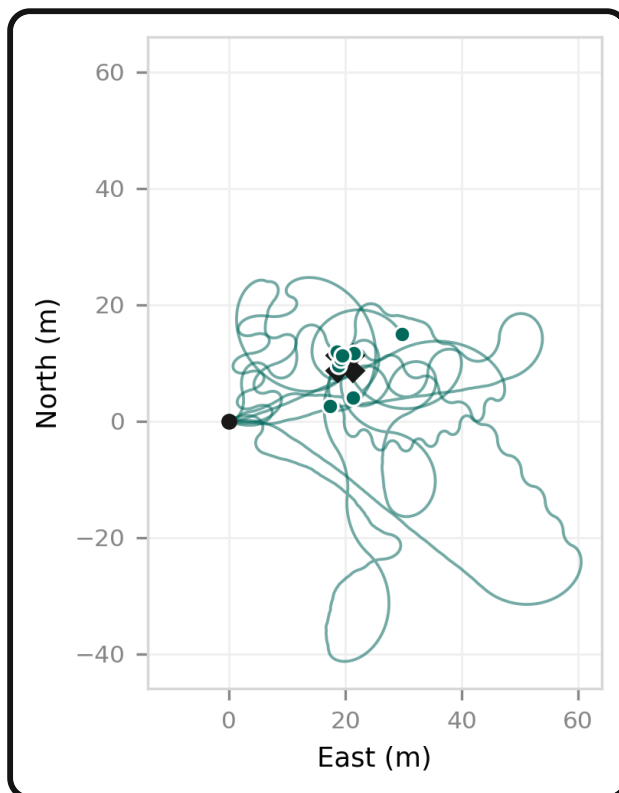
Identical conditions across rows. **Only the wind information available to the guidance differs.**

CONFIGURATION	MEDIAN MISS	90TH PCT
No estimate	8.57 m (6.49–12.46)	24.27 m
EKF onboard	3.89 m (2.85–5.28)	10.75 m
Told the true wind	4.62 m (3.37–5.44)	10.93 m

Intervals are 95% bootstrap percentile intervals on the median.



No estimate – touchdowns scattered.



EKF – touchdowns clustered on target.

Drop by drop, the EKF cuts the miss by a median **4.26 m** (95% CI 3.08–6.57) and wins **112 of 150**.

What this does not show. The EKF and true-wind intervals overlap, so at this sample size those two are *not* separated. Onboard estimation beats not knowing the wind; it is not shown to beat being told it.

SO → So build the most capable canopy you can afford?

THE ONE NUMBER THAT DECIDES EVERYTHING

No. A heavier one is worse.

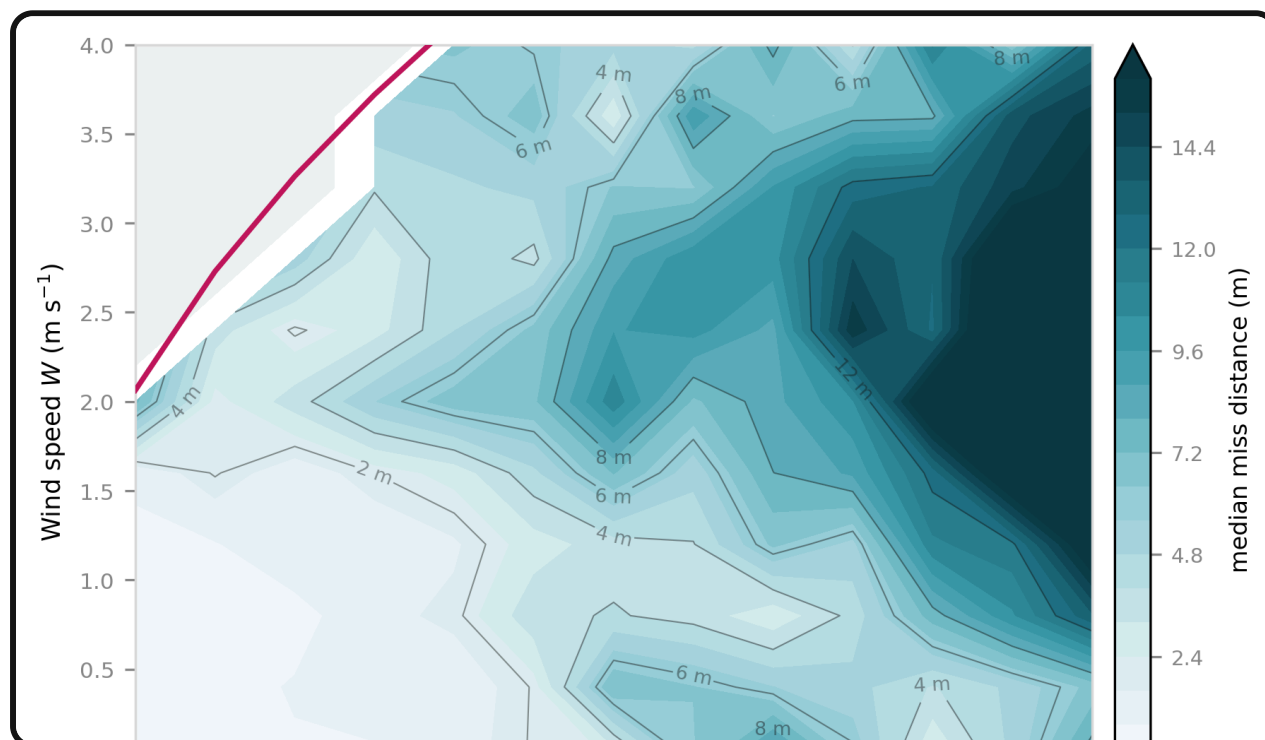
To make headway against wind W it must fly faster than W , forcing wing loading up as W^2 . But turn radius rises *linearly* with loading — and turn radius is the smallest correction it can make in the final seconds.

$$R_{\min} \propto W^2$$

The two requirements fight. The error floor follows the wind you *designed for*, not the wind you meet.

Building for a gale you never meet makes you less accurate on a calm day. The right choice is the lowest loading that still penetrates, with margin — not the most capable one.

Loading is set by **ballast**. Unlike canopy area, it can be matched to the forecast on the morning of the drop.



Every combination of loading and wind speed, 24 drops each. The dark valley is the best choice — and it moves with the wind.



SCAN 2 OF 2 · FINISH HERE

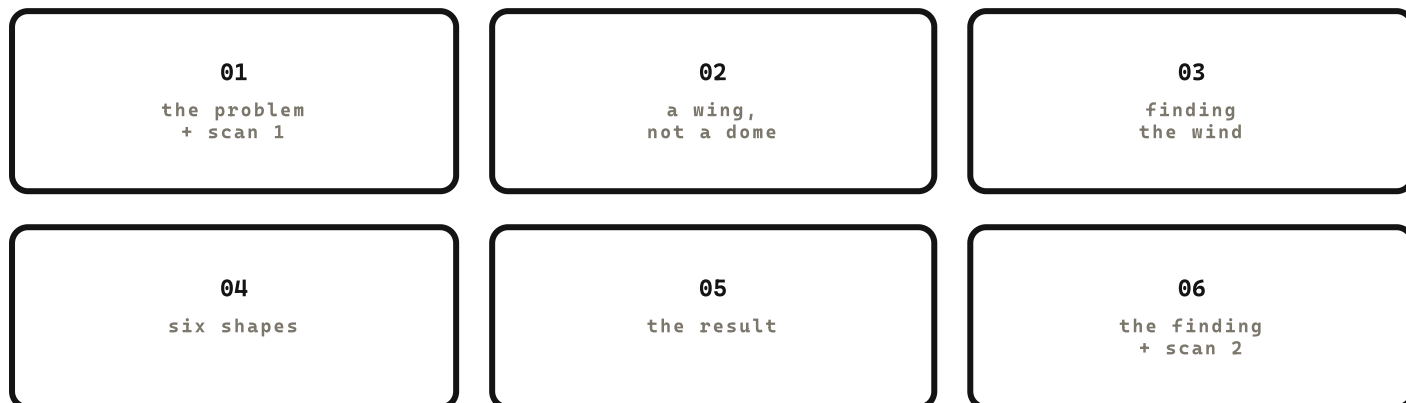
Now watch it land.

Same drop, same wind, flown with everything on this poster. The dotted line is where it went before.

✓ **23.9 m off, down to 2.35 m — at your feet.**

Sticking it together

Six sheets, three across and two down. Each panel is complete in itself, so **nothing needs trimming and nothing crosses a join**. The white border of each sheet becomes the gutter between panels.



1. Print pages 1–6 on A4 at **“Actual size” / 100%**, so the panels come out the size the layout was built for and the margins match. (The codes themselves were tested down to 70% and still scan — it is the layout that wants full size, not them.)
2. Lay all six out on the backing sheet *before* gluing anything. Line up the tops of panels 01–03, then 04–06.
3. Glue the middle sheet of the top row first and work outwards, so any error is shared between both sides instead of accumulating at one edge.
4. **Scan both QR codes with your own phone once the poster is up.** Test under the lighting in the hall, not at home.

If a QR code will not scan

Both codes were decoded from this very PDF at every resolution from draft to 300 dpi, so the artwork is not the problem. In practice it is nearly always **glare** — a reflection off tape, glue or a glossy sheet, sitting across the middle of the code. Move so the light is behind you rather than behind the phone. Failing that, low ink leaves gaps in the modules: reprint that one sheet. The codes carry the highest error correction level, so a crease or a fingerprint will not stop them.

Reading order

01 sets up the problem and sends them to the first scan. 02–04 are the model: why it can steer, how it finds the wind, how far it really has to fly. 05 is the evidence, including what it does *not* show. 06 is the finding, and sends them to the second scan — so a judge who only reads the top row still gets the problem, the mechanism and a demonstration.

Full paper and the interactive model: **1101-hub.github.io/guided-airdrop**